

$$\begin{aligned}
 \rightarrow \text{Recall} &= \frac{TP}{TP+FN} \\
 &= \frac{90}{90+210} \\
 &= \frac{399}{300} \\
 &= 1.33 \\
 &= 0.3
 \end{aligned}$$

$\rightarrow$ F1 Score

$$\begin{aligned}
 F &= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times 0.39 \times 0.3}{0.39 + 0.3} \\
 &= \frac{0.234}{0.69} = 33.9\%
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow \text{Accuracy: } & \frac{TP+TN}{TP+FN+FP+TN} \\
 &= \frac{90+9560}{90+210+140+9560} \\
 &= \frac{2650}{10000} \\
 &= 0.265 \\
 &= 26.5\%
 \end{aligned}$$

Answer

$$\begin{aligned}
 \rightarrow \text{Error Rate: } & 1 - \text{Accuracy} \\
 &= 1 - 0.265 \\
 &= 0.735
 \end{aligned}$$

Actual class / Predicted class	Cancer = yes	Cancer = no	total
Cancer = yes	90	210	300
Cancer = no	140	9560	9700
Total	230	9770	10000

A \ P	C	7C	
C	TP	FN	P
7C	FP	TN	N
	P'	N'	All.

Precision:

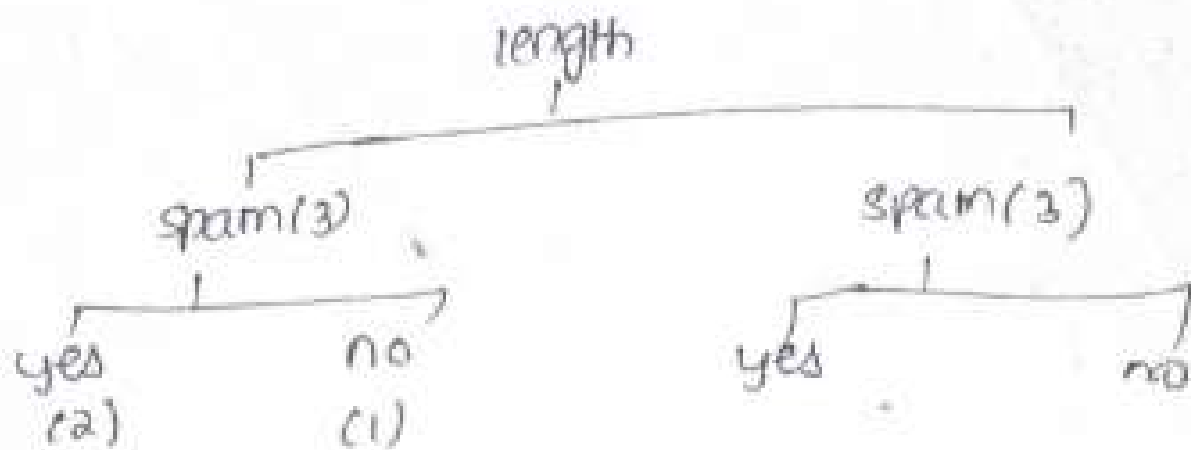
$$\frac{TP}{TP+FP} = \frac{90}{90+140} = \frac{90}{230} = 0.39 \times 100 = 39\%$$

specificity:

$$\frac{TN}{FP+TN} = \frac{9560}{140+9560} = \frac{9560}{9700} = 0.985$$

Sensitivity:

$$\frac{TP}{TP+FN} = \frac{90}{300} = 0.3$$



$$P(A|B) = P(\text{length=short} | \text{spam})$$

$$P(\text{length=short} | \text{spam}) = \frac{2}{3}$$

$$P(\text{length=short} | \text{not spam}) = \frac{1}{3}$$

$$P(L=short | \text{not spam}) = \frac{2}{3}$$

Feature	value	$P(\text{value} \text{spam})$	$P(\text{value} \text{not spam})$
offer	yes	$\frac{2}{3}$	$\frac{1}{3}$
	no	$\frac{1}{3}$	$\frac{2}{3}$
free	yes	$\frac{3}{3}$	$\frac{0}{3}$
	no	$\frac{0}{3}$	$\frac{3}{3}$
length	yes	$\frac{2}{3}$	$\frac{1}{3}$
	no	$\frac{1}{3}$	$\frac{2}{3}$

(39)

Classify the new Email?

New Email:

$x = (\text{offer} = \text{yes}, \text{free} = \text{no}, \text{length} = \text{short})$

using naive Bayes

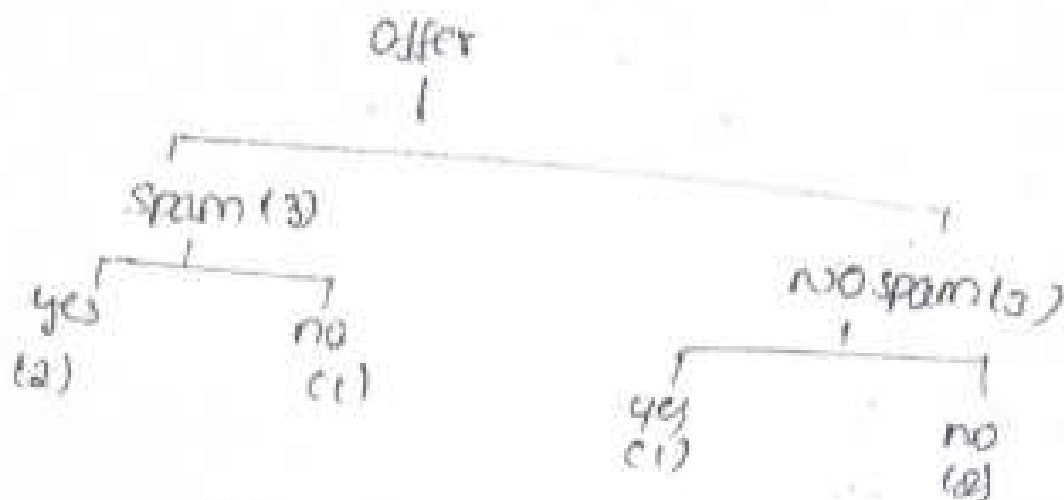
$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

$P(A)$ given mail = spam.

$$P(\text{spam} | x) \propto P$$

(i) Conditional probabilities;

offer



$$P(A|B) = \frac{P(\text{offer}=\text{yes})}{P(\text{spam})}$$

$$= \frac{2}{2+1} = \frac{2}{3}$$

$$P(\text{offer}=\text{yes}|\text{spam})$$

$$= 1/3$$

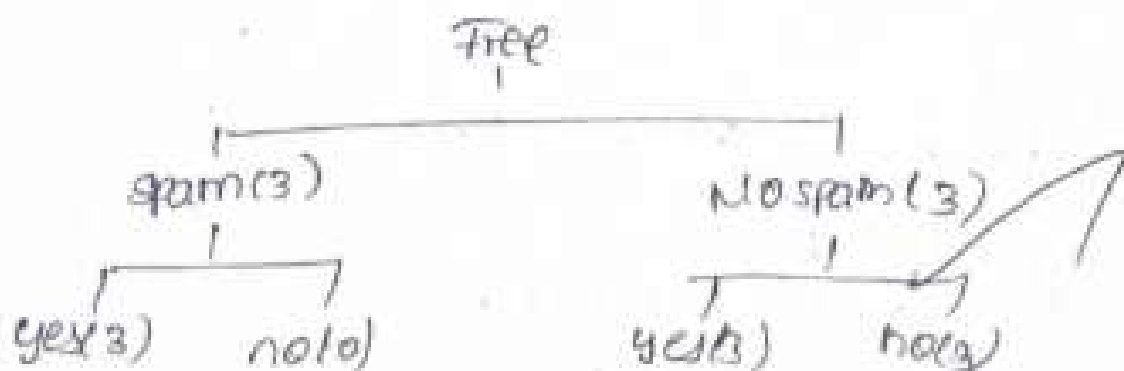
$$P(\text{offer}=\text{yes}|\text{no spam})$$

$$= 1/3$$

$$P(\text{offer}=\text{no}|\text{no spam})$$

$$= 2/3$$

Free :



$$P(A|B) = \frac{P(\text{offer}=\text{yes}|\text{spam})}{P(\text{spam})}$$

$$= 3/3 = 1$$

$$P(\text{offer}=\text{no}|\text{spam})$$

$$= 0/3 = 0$$

$$P(\text{offer}=\text{yes}|\text{no spam})$$

$$= 0/3 = 0$$

$$P(\text{offer}=\text{no}|\text{no spam})$$

$$= 3/3 = 1$$

RA) given mail = spam

$$\Rightarrow P(\text{spam} | x) \propto P(\text{spam}) \times P(\text{yes} | \text{spam}) \times P(\text{no} | \text{spam}) \\ \times P(\text{short} | \text{spam})$$

$$= 3/6 \times 2/3 \times 0/3 \times 2/3$$

$$= 0$$

PA) given mail = not spam

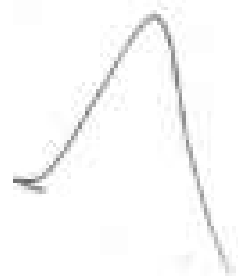
$$P(\text{not spam} | x) \propto P(\text{not spam}) \times P(\text{yes} | \text{not spam}) \times \\ P(\text{no} | \text{not spam}) \times P(\text{short} | \text{not spam})$$

$$= 3/6 \times 1/3 \times 3/3 \times 1/3$$

$$= 1/18$$

$$\therefore P(\text{not spam} | x) > P(\text{spam} | x)$$

$\therefore$ The new Email is classified as (not spam).



3) An email service provider wants to classify Emails as Spams or not-spam using features:

- Contains "offer" (yes/no)
- Contains "free" (yes/no)
- Email length (short/long)

Offer	free	length	Class.
yes	yes	short	spam
yes	no	long	not-spam
no	yes	short	spam
no	no	long	not-spam
yes	yes	long	spam
no	no	short	not-spam

(4) Prior probabilities

total Emails = 6

→ spam = 3

→ not-spam = 3

$$\Rightarrow P(\text{spam}) = 3/6 = 0.5$$

$$\Rightarrow P(\text{not}) = 3/6 = 0.5$$

A	P	C	-C	
	TP	FN		P
	25	20		45
C	FP	TN		N
-C	5	50		55
	P	N		All
	30	70		100

Total images = 100

Actual cat images = 30.

	Y	N	Total
	25	20	45
	5	50	55
	30	70	100

$$\text{precision} = \frac{TP}{TP+FP}$$

$$= \frac{25}{25+5}$$

$$= \frac{25}{30} = 0.83 = 83\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{25}{25+20} = \frac{25}{45} = 0.55 = 55\%$$

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 0.83 \times 0.55}{0.83 + 0.55}$$

$$= \frac{0.913}{1.38}$$

$$= 0.66 = 66\%$$

CO3 AT-2.

- ① A shepherd boy gets bored tending the ~~team's~~ flock to have some fun he cries out "wolf". Even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing joke on them. One night, the shepherd boy sees a

Act	Pre	
	Y	N
Y	TP Real wolf came, warning acted upon.	No wolf came FN false warning
N	wolf came but warning not FP acted upon.	TN no wolf no warning

real wolf approaching & the fooled again & stay in their houses. The hungry turns flock into lamb chops. Identifying TP, FP, FN, TN.

②

Assume there are 100 images, 30 of them depict a cat the rest, don't. A machine learning model predicts the occurrence of a cat

in a set of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat

images. Identify TP, FP, FN, TN calculate precision, recall & F1 score

Given,

Total images = 100

Actual cat images = 30

